

**Effects of Reward Volatility and Perceptual Alterations on Human Reinforcement
Learning**

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Introduction (Written by Dominic Jaccard, edited by Jane Liu)

Learning rate (α) is a measure of how quickly an actor updates its information after a reward prediction error. During voluntary choice, each time the actor receives an outcome it doesn't expect, it will update its internal state by an amount depending on its learning rate. This mechanism allows actors to accurately predict cues and alter expectations in the case that the environment changes.

Optimally, organisms should not keep their learning rate constant. Through modulating learning rate, agents aim to maximize future rewards (Behrens et al., 2007). Higher learning rates indicate future decisions are more affected by the previous outcome. On the other hand, lower learning rates suggest that decisions are less affected by past results. Behrens, et al., (2007) observed that volatility is represented by the blood-oxygen-level-dependent (BOLD) response of fMRI signal in the anterior cingulate cortex (ACC) region when the outcome of choices is observed. The ACC is a brain area in the middle of the cerebral cortex, frontal and above the corpus callosum. It is critical for updating the representation of decision values, working with dopaminergic neurons and the ventral striatum to represent prediction error signals (Samejima et al., 2005; Bayer & Glimcher, 2005; McCoy et al., 2003, Waelti, 2001). This supports the idea that learning rate in humans depends on how volatile we estimate the environment to be. Our study provides a blueprint for determining whether volatility affects learning rate.

Another commonly used parameter when investigating learning is inverse temperature (β) that examines behavioral strategies. The Softmax Decision Rule suggests that higher inverse temperature β indicates more exploitative behavior while lower β indicates more exploratory behavior. This behavior is determined by a signal to noise ratio mechanism; if the expected value of a given action is detectably higher than another option, then an exploitative option is selected (Feng et al. 2021). Otherwise, an exploratory option may be selected. The gain on this signal is stronger in volatile environments, where recent information is more sparse and valuable. Therefore, we expected a more volatile environment to lead to lower β values, with participants preferring more exploitative behavior.

Furthermore, perceptual changes unrelated to reward values may still impact reward-based decision making. Gershman & Niv (2010) argues that people must make inferences about sensory information to determine cause and effect relationships. By adding extraneous visual cues, we plan to examine whether participants engage in exploratory behavior to conclude the nature of these perceptual changes. This type of learning is called "representation learning," named as such because the actor is updating their representation of the task via visual cues and determining which features of the environment are salient predictors of reward.

By introducing perceptual confounds, we are recruiting the frontoparietal attention control network (Niv et al. 2015) to help evaluate relevant attentional cues. Whether this drive to determine which cues are relevant will overcome reward motivations is an open question. By observing if participants engage in more exploratory behavior after these changes, we hope to understand how exploration and exploitation dynamics are modulated by the orbitofrontal cortex (OFC) and ventral tegmental area (VTA) in unstable reward environments.

Based on previous findings (Behrens et al. 2007), we suggest that participants will modulate their learning rates as a function of environmental predictability in reward frequency and show different behavioral strategies (exploit/explore behaviors) according to the change in reward volatility. Reward volatility introduces an uncertain contextual structure that does not directly manipulate reward probabilities, but still affects learning rate and behavioral strategy choice. Thus, we hypothesized that, under a stable reward size and reward uncertainty (30% vs. 70%) situation, derived from a Rescorla–Wagner model fit to choice behavior, α for the volatile and attention conditions would be significantly higher relative to the stable condition. Additionally, we hypothesized that β for the volatile and attention conditions would be significantly lower (more explorative behaviors compared to exploitative behaviors) relative to the stable condition.

Methodology (Written by Jane Liu)

Participants

This study includes a sample of 34 volunteers to participate in the study, recruited through a convenience sample of other college students in the Madison area. Participants will perform an adapted version of the two-bandit task based on previous research paradigms (Rouault et al., 2019).

The Behavioral Two-Choice Bandits Task

The two choice bandits task (see Rouault et al., 2019; Figure 1A) was modified to fit our research focus. The behavioral task comprises three conditions: the stable condition, the volatile condition, and the attention condition. The attention condition is set to compare with the stable condition, aiming to investigate how unrelated perceptual changes influence reward-based decision making. Each of these conditions has 80 trials, adding up to a total of 240 trials. In every trial of any condition, one bandit leads to rewards more frequently than the other (frequency: 70% vs. 30%). The reward size is consistent across trials (reward size: 1). The reward percentage is only related to the shape of the two bandits. Bandit location and color of bandit did not predict reward probability. A time limit was not applied in this behavior task to eliminate the influence of urgency on learning rate and behavioral strategy. These three conditions are pseudo-randomly arranged to limit the potential interfering effect of familiarity with the task. Under the stable and attention conditions, the probability that the two bandits receive rewards switches about every 16 trials. Under volatile conditions, that probability switches about every 8 trials. Under stable and volatile conditions, the two bandits will be located on the left or right side. Under the attention condition, the bandits may be horizontally or vertically aligned. The color and shape of the bandits also change over time under attention conditions.

Before the task, participants are told to choose between two bandits under the condition that one choice has a higher likelihood of receiving reward and that only one of the indices is related to the probability of receiving rewards. They were also told that their goal is to get as many rewards as possible.

Statistical Analysis

Learning rate (α) is calculated using the Rescorla–Wagner model

$$Q_{(t+1)}(\alpha_t) = Q_{(t)}(\alpha_t) + \alpha \cdot \delta_t.$$
 Inverse temperature (β) is calculated through the

Softmax Decision Rule $P_a = \frac{e^{\beta Q_a}}{e^{\beta Q_a} + e^{\beta Q_b}}$. Calculations and model fits were done in Python.

We used a mixed effect regression model to examine significant relationships between conditions and block orders. Because our study was administered within-subjects, the particular model was specifically chosen to control for individual differences between participants. Beta values greater than 20 were treated as erroneous model fits and those values along with the corresponding alpha fits were excluded from analyses.

Figures and Results (Written by Anders Kvalsvik, edited by Jane Liu)

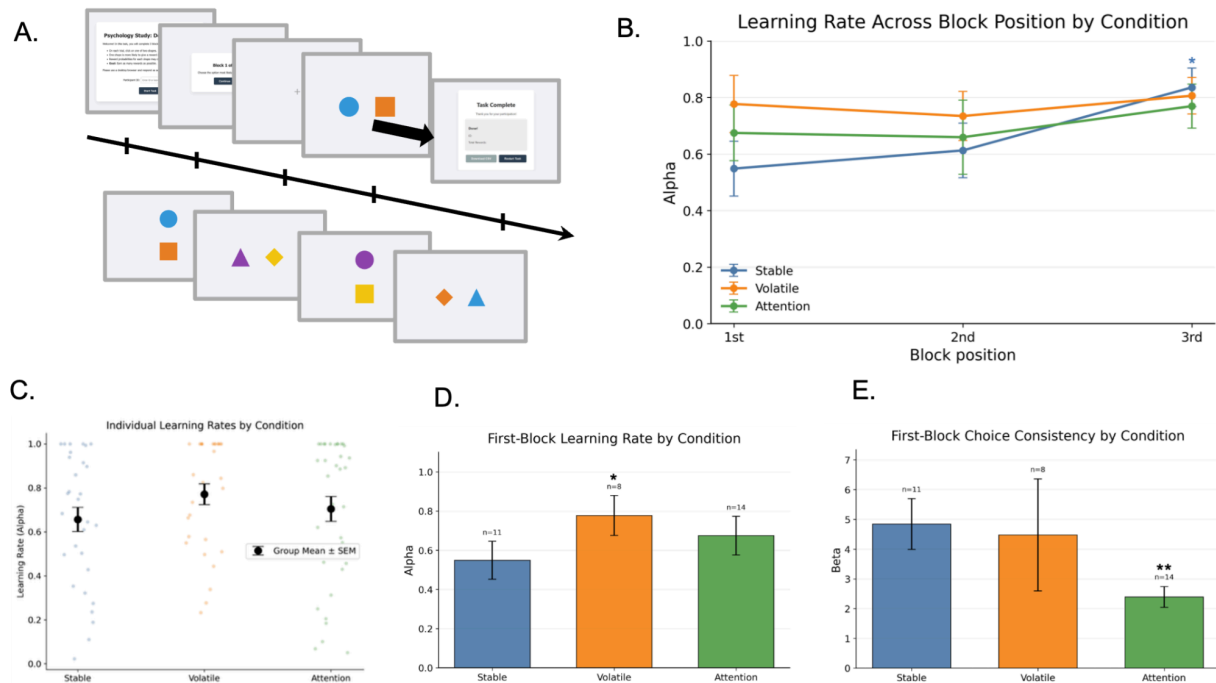


Figure 1. A. A simplified demonstration of the experiment setup (top panel: The overall setup; bottom panel: Four sets of options used in attention conditions.) B. Line chart depicts learning rate (α) across block position by condition. Error bars are standard error of mean. All conditions show learning rate increase as block position increases, with that of stable condition being the sharpest ($b=5.522$, $p_{\text{stable, block1-block3}}=0.030$). C. Scatterplot depicts learning rate (α) by conditions ($b_{\text{stable-volatile}}=3.112$, $p_{\text{stable-volatile}}=0.028$, $p_{\text{stable-attention}}=0.302$). Error bars are standard error of mean. D. Bar graph depicts learning rate (α) by conditions, including only the first blocks ($b_{\text{block1, stable-volatile}}=6.405$, $p_{\text{block1, stable-volatile}}=0.015$). Error bars are standard error of mean. E. Bar graph depicts inverse temperature (β) by condition, including only the first block ($b_{\text{block1, stable-attention}}=-0.798$, $p_{\text{block1, stable-volatile}}=0.006$). Error bars are standard error of mean. * $p < 0.05$, ** $p < 0.01$ vs stable conditions

Learning Rate (α)

In line with our initial hypotheses and existing literature, a main-effects mixed model confirmed that the estimated learning rate (α) was significantly higher in the volatile condition than in the stable condition ($b=3.112$, $p=0.028$, Table X, Figure 1C).

The difference in learning rate between the attention and stable conditions was not statistically significant ($p=0.302$, Table X, Figure 1C).

Subsequent interaction models revealed that block position significantly modulated the effect of condition on learning rate. Surprisingly, learning rates generally increased over time across all conditions. This escalation was sharpest in the stable condition, which yielded a significantly higher α when presented third compared to first ($b=5.522$, $p=0.030$, Table Y, Figure 1B). Furthermore, the disparity between volatile and stable blocks was most pronounced during the first block, where the volatile condition produced a significantly higher learning rate relative to the stable condition ($b=6.405$, $p=0.015$, Table Y, Figure 1D). However, a trending interaction indicated that this condition-driven effect weakened by the time the volatile block appeared in the third position, where there was less of a discrepancy ($p=0.082$).

Inverse Temperature (β)

The main-effects model did not reveal overall significant differences in inverse temperature for either the volatile ($p = 0.236$) or attention ($p = 0.087$) conditions when compared to the stable condition. However, interaction models showed that block position played a critical role in exploration versus exploitation dynamics. When analyzing only the first block, the attention condition was associated with a significantly lower β than the stable condition ($b = -0.798$, $p = 0.006$; Table Y, Figure 1E). This attention-related reduction in β was significantly attenuated when the attention condition was encountered in the second block ($b = 0.965$, $p = 0.040$), with a similar trend if presented third ($p = 0.092$). Finally, stable blocks exhibited a significantly lower inverse temperature parameter when presented third versus first ($b = -0.635$, $p = 0.044$, Table Y).

Summary

Overall the learning rates in the volatile condition were elevated relative to the stable condition, with this effect being amplified when the blocks were presented first. Furthermore, inverse temperature (β) was reduced for the attention condition relative to the stable condition when presented first. However, there are some limitations to these interpretations, as the log-likelihood values indicate better fits for inverse temperature (β) than for learning rate (α), suggesting greater accuracy for inverse temperature estimations. While these results are promising and certainly support the need for further exploration, future iterations of this study would likely benefit from greater experimental control and modeling rigor.

Table X Main-Effects Mixed Models Predicting Learning Rate and Inverse Temperature

Predictor	Outcome	Estimate	Test Stat (z)	95% CI	p-value
Intercept	logit(alpha)	2.150†	1.645	[-0.411, 4.711]	.100
Volatile vs. Stable	logit(alpha)	3.112*	2.197	[0.336, 5.888]	.028
Attention vs. Stable	logit(alpha)	1.401	1.032	[-1.261, 4.064]	.302
Block position 2 vs. 1	logit(alpha)	0.225	0.161	[-2.506, 2.955]	.872
Block position 3 vs. 1	logit(alpha)	1.530	1.130	[-1.123, 4.183]	.258
Intercept	log(beta)	1.211***	7.211	[0.882, 1.540]	< .001
Volatile vs. Stable	log(beta)	-0.206	-1.184	[-0.548, 0.135]	.236
Attention vs. Stable	log(beta)	-0.285†	-1.709	[-0.613, 0.042]	.087
Block position 2 vs. 1	log(beta)	0.007	0.039	[-0.329, 0.342]	.969
Block position 3 vs. 1	log(beta)	-0.120	-0.723	[-0.446, 0.206]	.470

Note. † $p \leq .10$, * $p < .05$, *** $p < .001$. Estimates were derived from Mixed-Effects Maximum Likelihood (ML) regressions. Model Log-likelihoods: logit(alpha) = -295.650; log(beta) = -105.129. Effect sizes represented by "Estimate" value.

Table Y Selected By-Block-Position Results From Interaction Models

Effect	Outcome	Estimate	Test Stat (z)	95% CI	p-value
Volatile vs. Stable at block position 1	logit(alpha)	6.405*	2.422	[1.223, 11.587]	.015
Attention vs. Stable at block position 1	logit(alpha)	3.851†	1.680	[-0.640, 8.343]	.093
Block position 3 vs. 1 within Stable	logit(alpha)	5.522*	2.166	[0.525, 10.519]	.030
Volatile × Block position 3	logit(alpha)	-6.607†	-1.740	[-14.049, 0.835]	.082
Attention vs. Stable at block position 1	log(beta)	-0.798**	-2.775	[-1.361, -0.234]	.006
Block position 3 vs. 1 within Stable	log(beta)	-0.635*	-2.012	[-1.253, -0.016]	.044

Note. † $p < .10$, * $p < .05$, ** $p < .01$. Estimates were derived from a Mixed-Effects Maximum Likelihood (ML) regression. Effect sizes represented by "Estimate" value.

Discussion (Written by Dominic Jaccard, edited by Anders Kvalsik and Jane Liu) .

Learning Rate

Fitted learning rate values were generally lower for the stable condition, higher for the attention condition, and highest for the volatile condition. However, only the difference between the stable and volatile condition is statistically significant ($n=34$, $b=3.112$, $p_{\text{volatile-stable}}=0.028$), with the volatile condition having a significantly higher α than the stable condition. Difference in learning rate between volatile and stable conditions is more extreme when only including the first block ($n=19$, $b=6.405$, $p_{\text{block1, volatile-stable}}=0.015$), as α values had mostly stabilized around a high value by the third block. Differences between attention conditions and stable conditions, either including data from all three blocks or only analyzing that from block 1, are close to but not statistically significant. We infer that the extreme α values we saw in our data close to 1 indicate that participants had developed a strategy by the end of the experiment wherein they selected one bandit until a reward was not administered, in which case they would switch to another bandit. Regardless of the order of conditions, most participants were behaving this way by the third block.

Although this behavior is not what we expected to happen, it was a consistent pattern presented in the results that contradicted the literature, and does reveal something about the participants. Comparing our experimental setup to Behrens (2007) we can see that the stable environment for their bandit task involved 120 trials, and their volatile environment swapped reward probabilities every 30 or 40 trials. Even our stable environment, which consisted of 16 trials, was much shorter than Behren's volatile environment. While we hypothesized that the results wouldn't differ, given that one trial was more volatile than the other, it appears that both trials contained a level of volatility too high for any cue to be conclusively disregarded. This means that, for many, the single previous cue was maximally relevant, leading to the "high- α " behavior described earlier.

We also observed a statistically significant difference in the fitted learning rates in stable conditions when it is presented as the first block compared to the third block ($b=5.522$, $p_{\text{stable, block1-3}} = 0.03$). This indicates that as participants became more familiar with the task, they seem to have developed a strategy of higher learning rates, which aligns with our hypothesis inferred from the relative extremity of their α values.

Inverse Temperature

We did find significant results for inverse temperature when only including the first block. The β value associated with the attention block was significantly lower than the stable block, even with a smaller sample size. This partially supports our hypothesis— we expected to find lower β values for the attention block compared to the stable block, but we also expected to find lower β values for the volatile block compared to the stable block, which we did not find.

The effect of the attention block on the participants depended on the block position. In the first block, the attention condition had the expected effect of visually indicating a volatile environment, which prompted more exploratory behavior. However, this behavior was not apparent in the second and third blocks. In completing the first

two blocks, participants may have been familiarized with the task paradigm, which would have allowed them to more reliably disregard the extraneous visual cues. This would suggest that people with more familiarity with an environment are less likely to change their behavior due to factors outside of those reward frequencies. Martens, 2007 supports this conclusion: they explore drivers' responses to changes to road signs, and find that people more familiar with a given driving route respond less to new road signs placed on this route. We suggest further study to explore how these effects extend to reward-based environments.

Neuroscience Extensions

It appears that the frontoparietal attention control network was active during this task (Niv, 2015), given that participants quickly filtered out irrelevant attentional cues, especially in the later blocks. We suggest that further research is needed to investigate the interactions between attention and reward, specifically questioning what factors influence attentional filtering. For example, a further experiment could explore two variations on the attention condition: one with a superficial reward (points) and one with a meaningful reward (something more appealing to the participant, like candy or a gift card). In this instance, the orbitofrontal cortex may have more weight, as participants may be more inclined to get a meaningful reward and more likely to filter out attentional cues. These explorations of attentional control make for a natural extension of our research especially suited to studies focusing on neural recording methodologies including fMRI and MEG. Non-invasive neuromodulating techniques, such as TMS and deep-brain non-invasive temporal interference electrical stimulation, may also be used in future research to investigate causal relationships.

Limitations

There are several limitations present in the study that prevent its generalizability. Demand characteristics were almost certainly present on several fronts. Most participants were recruited from a convenience sample of friends, family, and other close contacts, meaning that many participants had exterior motivations to perform in a way that satisfied the research team's interests. Additionally, our task was quite long, meaning that participants may have experienced attrition by the end. This would also explain why the "high- α " strategy developed in the later blocks, as participants may have resorted to a deterministic strategy to minimize effort as the experiment progressed. In a future experiment, in addition to finding a more broadly representative sample, a small "break" activity should be implemented between blocks to ensure that participants do not experience attrition effects.

Despite our interesting findings on the explore-exploit behavior dynamics, we still must note that we removed a significant number of outliers and data points, meaning that only parts of our participants' behaviors supported our hypotheses. The results for inverse temperature were also less robust than the learning rate results, as we did not find any significant differences between learning rate between conditions. Again, the "high- α " behavioral pattern contributed to this result. Because β can be any value above zero, with exploitative behavior increasing the β value, we saw extremely high β values, with some above 30, far higher than the median. A larger sample size may lead to better results, and we encourage modifications to the experimental paradigm to decrease volatility overall in future studies.

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Methods: J. Liu

Results: A. Kvalsik (primary), J. Liu (secondary & graphs)

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References: All members